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R11954346

Rankine Cycle

Problem 1.1, 2.1, 3

1 — Import Libraries

```
In [109... pip install numpy matplotlib CoolProp pyromat tespy
```

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
,Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
,Requirement already satisfied: CoolProp in /usr/local/lib/python3.12/dist-packages (7.2.0)
,Requirement already satisfied: pyromat in /usr/local/lib/python3.12/dist-packages (2.2.6)
,Requirement already satisfied: tespy in /usr/local/lib/python3.12/dist-packages (0.9.15.post1)
,Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
,Requirement already satisfied: cyclor>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
,Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
,Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
,Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
,Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
,Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
,Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
,Requirement already satisfied: fluprodia>=4.1.2 in /usr/local/lib/python3.12/dist-packages (from tespy) (4.1.2)
,Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from tespy) (3.1.6)
,Requirement already satisfied: pandas>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from tespy) (2.2.2)
,Requirement already satisfied: pint in /usr/local/lib/python3.12/dist-packages (from tespy) (0.25.3)
,Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from tespy) (1.16.3)
,Requirement already satisfied: tabulate>=0.8.2 in /usr/local/lib/python3.12/dist-packages (from tespy) (0.9.0)
,Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.0->tespy) (2025.2)
,Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.0->tespy) (2026.1)
,Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
,Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->tespy) (3.0.3)
,Requirement already satisfied: flexcache>=0.3 in /usr/local/lib/python3.12/dist-packages (from pint->tespy) (0.3)
,Requirement already satisfied: flexparser>=0.4 in /usr/local/lib/python3.12/dist-packages (from pint->tespy) (0.4)
,Requirement already satisfied: platformdirs>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from pint->tespy) (4.9.6)
,Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from pint->tespy) (4.15.0)

```

In [110... import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from matplotlib.patches import FancyArrowPatch
from CoolProp.CoolProp import PropsSI

plt.rcParams.update({
    'font.size': 11,
    'axes.titlesize': 13,
    'axes.labelsize': 12,
    'legend.fontsize': 10,
    'figure.dpi': 120
})

def get_saturation_dome(fluid='Water', n=400):
    """
    Returns sat. liquid (Q=0) and sat. vapor (Q=1) curve data.
    Used as the background dome on T-s, P-h, and h-s diagrams.
    """
    T_min = PropsSI('Tmin', fluid) + 1
    T_crit = PropsSI('Tcrit', fluid)
    T_arr = np.linspace(T_min, T_crit - 0.05, n)

    def prop_arr(prop, Q):
        return np.array([PropsSI(prop, 'T', T, 'Q', Q, fluid) for T in T_arr])

    return {
        'T': T_arr - 273.15,          # °C
        's_f': prop_arr('S', 0)/1e3, # kJ/(kg·K)
        's_g': prop_arr('S', 1)/1e3,
        'h_f': prop_arr('H', 0)/1e3, # kJ/kg
        'h_g': prop_arr('H', 1)/1e3,
        'P': prop_arr('P', 0)/1e3,   # kPa
    }

dome = get_saturation_dome()
print("✅ Libraries imported and saturation dome data ready")

```

✅ Libraries imported and saturation dome data ready

Problem 1.1 — Basic Rankine Cycle

Problem Conditions

- Working fluid: **Water (Steam)**
- Turbine inlet: **500°C, 1.4 MPa** (superheated vapor)
- Condenser pressure: **0.01 MPa** (10 kPa)
- Turbine isentropic efficiency: $\eta_t = 90\%$
- Pump isentropic efficiency: $\eta_p = 80\%$

State Point Numbers

State 5, sat L → (Pump) → State 6 → (Boiler) → State 3 → (Turbine) → State 4 → (Condenser) → State 5

State	Location	Description
3	Turbine inlet	Superheated vapor (supV)
4'	Ideal turbine outlet	Isentropic assumption
4	Actual turbine outlet	With $\eta_t = 90\%$
5, sat L	Condenser outlet (pump inlet)	Saturated liquid (Q=0)
6	Pump outlet (boiler inlet)	Compressed liquid (compL)

2 — Set Conditions

```
In [111]: fluid = 'Water'

P_high = 1.4e6
P_low  = 0.01e6

T3 = 500 + 273.15

eta_t = 0.90
eta_p = 0.80
```

```

print(f"Boiler pressure      P_high = {P_high/1e6:.2f} MPa")
print(f"Condenser pressure P_low  = {P_low/1e6:.3f} MPa")
print(f"Turbine inlet temp T3     = {T3-273.15:.1f} °C")
print(f"Turbine efficiency  $\eta_t$    = {eta_t*100:.0f}%")
print(f"Pump efficiency  $\eta_p$     = {eta_p*100:.0f}%")

```

```

Boiler pressure      P_high = 1.40 MPa
,Condenser pressure P_low  = 0.010 MPa
,Turbine inlet temp T3     = 500.0 °C
,Turbine efficiency  $\eta_t$    = 90%
,Pump efficiency  $\eta_p$     = 80%

```

3 — State Point Property Calculation

```

In [112... h3 = PropsSI('H', 'T', T3, 'P', P_high, fluid)
s3 = PropsSI('S', 'T', T3, 'P', P_high, fluid)

s4_prime = s3
h4_prime = PropsSI('H', 'P', P_low, 'S', s4_prime, fluid)
T4_prime = PropsSI('T', 'P', P_low, 'S', s4_prime, fluid)
x4_prime = PropsSI('Q', 'P', P_low, 'S', s4_prime, fluid)

h4 = h3 - eta_t * (h3 - h4_prime)
s4 = PropsSI('S', 'P', P_low, 'H', h4, fluid)
T4 = PropsSI('T', 'P', P_low, 'H', h4, fluid)
x4 = PropsSI('Q', 'P', P_low, 'H', h4, fluid)

h5 = PropsSI('H', 'P', P_low, 'Q', 0, fluid)
s5 = PropsSI('S', 'P', P_low, 'Q', 0, fluid)
T5 = PropsSI('T', 'P', P_low, 'Q', 0, fluid)
v5 = 1 / PropsSI('D', 'P', P_low, 'Q', 0, fluid)

W_pump_ideal = v5 * (P_high - P_low)
W_pump_actual = W_pump_ideal / eta_p

h6 = h5 + W_pump_actual
s6 = PropsSI('S', 'P', P_high, 'H', h6, fluid)
T6 = PropsSI('T', 'P', P_high, 'H', h6, fluid)

print("=" * 72)
print("  Problem 1.1 – State Point Property Table")
print("=" * 72)
print(f"{'State':^12} | {'T [°C]':^10} | {'P [MPa]':^9} | {'q':^8} | {'H [kJ/k}
print("-" * 72)

rows = [
    ("3 (Turb.in)", T3, P_high, 'supV', h3, s3),
    ("4' (Ideal)", T4_prime, P_low, x4_prime, h4_prime, s4_prime),
    ("4 (Actual)", T4, P_low, x4, h4, s4),
    ("5, sat L", T5, P_low, 0.0, h5, s5),
    ("6 (Pump out)", T6, P_high, 'compL', h6, s6),

```

```
]

for name, T, P, q, h, s in rows:
    q_str = f"{q:.3f}" if isinstance(q, float) else q
    print(f"{name:^12} | {T-273.15:^10.2f} | {P/1e6:^9.3f} | {q_str:^8} | {h/1

print("=" * 72)
print(f"\n Specific volume v5 = {v5*1e6:.0f} cm³/kg  ({v5:.5f} m³/kg)")
```

```
=====
, Problem 1.1 – State Point Property Table
,=====
, State | T [°C] | P [MPa] | q | H [kJ/kg] | S [kJ/kgK]
,-----
, 3 (Turb.in) | 500.00 | 1.400 | supV | 3474.78 | 7.6047
, 4' (Ideal) | 45.81 | 0.010 | 0.927 | 2410.30 | 7.6047
, 4 (Actual) | 45.81 | 0.010 | 0.972 | 2516.75 | 7.9384
, 5, sat L | 45.81 | 0.010 | 0.000 | 191.81 | 0.6492
, 6 (Pump out) | 45.94 | 1.400 | compL | 193.56 | 0.6503
,=====
,
, Specific volume v5 = 1010 cm³/kg (0.00101 m³/kg)
```

4 — Energy Calculations

```
In [113... W_turbine = (h3 - h4) / 1e3

W_pump = (h6 - h5) / 1e3
Q_boiler = (h3 - h6) / 1e3
Q_cond = (h4 - h5) / 1e3
W_net = W_turbine - W_pump
eta_th = W_net / Q_boiler
BWR = W_pump / W_turbine

Power_target = 1e6
m_dot = Power_target / (W_net * 1e3)
m_dot_hr = m_dot * 3600

print("=" * 48)
print(" Problem 1.1 – Energy Calculation Results")
print("=" * 48)
print(f" (a) Turbine work output W_turbine = {W_turbine:.1f} kJ/kg")
print()
print(f" (b) Pump work input W_pump = {W_pump:.2f} kJ/kg")
print(f" Boiler heat input Q_boiler = {Q_boiler:.1f} kJ/kg")
print(f" Condenser heat reject. Q_cond = {Q_cond:.1f} kJ/kg")
print(f" Net work output W_net = {W_net:.1f} kJ/kg")
print(f" Thermal efficiency η_th = {eta_th:.4f} ({eta_th*100:.2
print(f" Back Work Ratio BWR = {BWR*100:.2f}%")
print()
print(f" (c) Circulation rate (1 MW) ṁ = {m_dot:.4f} kg/s")
```

```

print(f"                                     ṁ = {m_dot_hr:.0f} kg/h")
print("=" * 48)

```

```

=====
,   Problem 1.1 – Energy Calculation Results
=====
,   (a) Turbine work output    W_turbine = 958.0 kJ/kg
,
,   (b) Pump work input       W_pump    = 1.76 kJ/kg
,       Boiler heat input     Q_boiler  = 3281.2 kJ/kg
,       Condenser heat reject. Q_cond   = 2324.9 kJ/kg
,       Net work output       W_net     = 956.3 kJ/kg
,       Thermal efficiency    η_th     = 0.2914 (29.14%)
,       Back Work Ratio       BWR      = 0.18%
,
,   (c) Circulation rate (1 MW) ṁ = 1.0457 kg/s
,                                   ṁ = 3765 kg/h
=====

```

6 — Visualization: T-s Diagram

```

In [114... fig, ax = plt.subplots(figsize=(9, 6))

ax.plot(dome['s_f'], dome['T'], 'steelblue', lw=2, label='Saturation Dome')
ax.plot(dome['s_g'], dome['T'], 'steelblue', lw=2)
ax.fill_betweenx(dome['T'], dome['s_f'], dome['s_g'], alpha=0.06, color='steelblue')

h6_ideal = h5 + W_pump_ideal
s6_ideal = PropsSI('S', 'P', P_high, 'H', h6_ideal, fluid)
s_id = [s5, s6_ideal, s3, s4_prime, s5]
T_id = [T5, PropsSI('T', 'P', P_high, 'H', h6_ideal, fluid), T3, T4_prime, T5]
ax.plot([s/1e3 for s in s_id], [T-273.15 for T in T_id],
        'g--', lw=1.8, alpha=0.7, label='Ideal Cycle (η=100%)', zorder=4)

s_act = [s5, s6, s3, s4, s5]
T_act = [T5, T6, T3, T4, T5]
ax.plot([s/1e3 for s in s_act], [T-273.15 for T in T_act],
        'tomato', lw=2.2, marker='o', ms=8, label='Actual Cycle (η_t=85%, η_p=85%)', zorder=5)

pt_labels = [
    (s5/1e3, T5-273.15, '5\n(sat L)', (-0.35, -15)),
    (s6/1e3, T6-273.15, '6', (-0.15, +12)),
    (s3/1e3, T3-273.15, '3', (+0.05, +8)),
    (s4/1e3, T4-273.15, '4', (+0.08, -18)),
    (s4_prime/1e3, T4_prime-273.15, "4'", (-0.35, +8)),
]
for sx, Tx, lbl, (dsx, dTx) in pt_labels:
    ax.annotate(lbl, xy=(sx, Tx), xytext=(sx+dsx, Tx+dTx),
                fontsize=10, fontweight='bold', color='darkred',
                arrowprops=dict(arrowstyle='->', color='gray', lw=1))

```

```

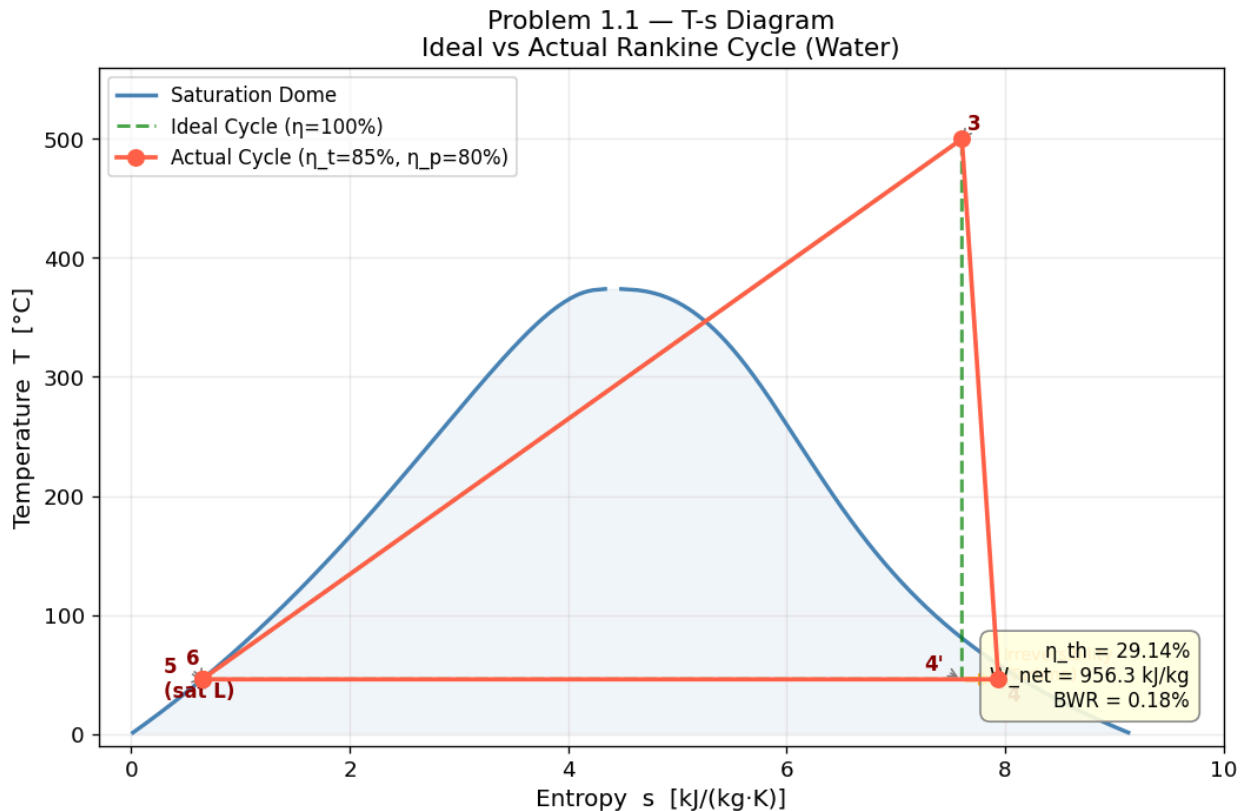
ax.annotate('', xy=(s4/1e3, T4-273.15), xytext=(s4_prime/1e3, T4_prime-273.15),
            arrowprops=dict(arrowstyle='->', color='orange', lw=2.5))
ax.text(s4/1e3+0.05, (T4+T4_prime)/2-273.15, 'Irreversibility\n(Turbine)',
        fontsize=8.5, color='darkorange')

ax.set_xlabel('Entropy s [kJ/(kg·K)]')
ax.set_ylabel('Temperature T [°C]')
ax.set_title('Problem 1.1 – T-s Diagram\nIdeal vs Actual Rankine Cycle (Water)')
ax.legend(loc='upper left')
ax.set_xlim([-0.3, 10])
ax.set_ylim([-10, 560])
ax.grid(True, alpha=0.25)

textstr = f'η_th = {eta_th*100:.2f}%\nW_net = {W_net:.1f} kJ/kg\nBWR = {BWR*100:.2f}%'
ax.text(0.97, 0.05, textstr, transform=ax.transAxes, fontsize=10,
        verticalalignment='bottom', horizontalalignment='right',
        bbox=dict(boxstyle='round,pad=0.5', facecolor='lightyellow', edgecolor='black'))

plt.tight_layout()
plt.savefig('ex51_ts_diagram.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: ex51_ts_diagram.png")

```



Saved: ex51_ts_diagram.png

Problem 2.1 — Rankine Cycle with Reheat

Problem Conditions

- 1st turbine inlet: **500°C, 6 MPa**
- 1st turbine outlet = reheater inlet: pressure **1.4 MPa**
- After reheat, 2nd turbine inlet: **500°C, 1.4 MPa** (same as Problem 1.1 State 3)
- 2nd turbine outlet = condenser: **0.01 MPa**
- Pump: **0.01 MPa → 6 MPa**
- $\eta_t = 90\%$, $\eta_p = 80\%$

State Point Numbers

State 7 (sat L) → (Pump) → State 8 → (Boiler) → State 3
→ (Turbine I) → State 4 → (Reheater) → State 5
→ (Turbine II) → State 6 → (Condenser) → State 7

```
In [66]: fluid = 'Water'

P_high2 = 6e6
P_mid   = 1.4e6
P_low2  = 0.01e6

T_turb1_in = 500 + 273.15
T_turb2_in = 500 + 273.15

eta_t2 = 0.9
eta_p2 = 0.8

print(f"1st turbine inlet : {T_turb1_in-273.15:.0f}°C, {P_high2/1e6:.0f} MPa")
print(f"Reheater pressure : {P_mid/1e6:.1f} MPa")
print(f"2nd turbine inlet : {T_turb2_in-273.15:.0f}°C, {P_mid/1e6:.1f} MPa")
print(f"Condenser pressure: {P_low2/1e6:.3f} MPa")
```

1st turbine inlet : 500°C, 6 MPa
, Reheater pressure : 1.4 MPa
, 2nd turbine inlet : 500°C, 1.4 MPa
, Condenser pressure: 0.010 MPa

```
In [115... h3_r = PropsSI('H', 'T', T_turb1_in, 'P', P_high2, fluid)
s3_r = PropsSI('S', 'T', T_turb1_in, 'P', P_high2, fluid)
T3_r = T_turb1_in

s4p_r = s3_r
h4p_r = PropsSI('H', 'P', P_mid, 'S', s4p_r, fluid)
T4p_r = PropsSI('T', 'P', P_mid, 'S', s4p_r, fluid)

W_turb1_ideal = h3_r - h4p_r
W_turb1_actual = eta_t2 * W_turb1_ideal
h4_r = h3_r - W_turb1_actual
s4_r = PropsSI('S', 'P', P_mid, 'H', h4_r, fluid)
T4_r = PropsSI('T', 'P', P_mid, 'H', h4_r, fluid)

h5_r = PropsSI('H', 'T', T_turb2_in, 'P', P_mid, fluid)
s5_r = PropsSI('S', 'T', T_turb2_in, 'P', P_mid, fluid)
T5_r = T_turb2_in
Q_reheat = (h5_r - h4_r) / 1e3

s6p_r = s5_r
h6p_r = PropsSI('H', 'P', P_low2, 'S', s6p_r, fluid)

W_turb2_ideal = h5_r - h6p_r
W_turb2_actual = eta_t2 * W_turb2_ideal
h6_r = h5_r - W_turb2_actual
s6_r = PropsSI('S', 'P', P_low2, 'H', h6_r, fluid)
T6_r = PropsSI('T', 'P', P_low2, 'H', h6_r, fluid)
x6_r = PropsSI('Q', 'P', P_low2, 'H', h6_r, fluid)

h7_r = PropsSI('H', 'P', P_low2, 'Q', 0, fluid)
s7_r = PropsSI('S', 'P', P_low2, 'Q', 0, fluid)
T7_r = PropsSI('T', 'P', P_low2, 'Q', 0, fluid)
v7_r = 1 / PropsSI('D', 'P', P_low2, 'Q', 0, fluid)

W_pump2_ideal = v7_r * (P_high2 - P_low2)
W_pump2_actual = W_pump2_ideal / eta_p2
h8_r = h7_r + W_pump2_actual
s8_r = PropsSI('S', 'P', P_high2, 'H', h8_r, fluid)
T8_r = PropsSI('T', 'P', P_high2, 'H', h8_r, fluid)

W_turb_total = (W_turb1_actual + W_turb2_actual) / 1e3
```

```

W_pump2_kJ      = W_pump2_actual / 1e3
W_net2          = W_turb_total - W_pump2_kJ
Q_boiler2       = (h3_r - h8_r) / 1e3
Q_total         = Q_boiler2 + Q_reheat
eta_th2         = W_net2 / Q_total
m_dot2          = 1e6 / (W_net2 * 1e3) * 3600

print("=" * 70)
print("  Problem 2.1 – State Point Property Table")
print("=" * 70)
print(f"{'State':^14} | {'T [°C]':^9} | {'P [MPa]':^9} | {'H [kJ/kg]':^11} | {'S [kJ/kg·K]':^11}")
print("-" * 70)
rows2 = [
    ("3 (Turb.I in)", T3_r, P_high2, h3_r, s3_r),
    ("4' (I ideal)", T4p_r, P_mid, h4p_r, s4p_r),
    ("4 (Turb.I out)", T4_r, P_mid, h4_r, s4_r),
    ("5 (Reheat out)", T5_r, P_mid, h5_r, s5_r),
    ("6 (Turb.II out)", T6_r, P_low2, h6_r, s6_r),
    ("7 sat L", T7_r, P_low2, h7_r, s7_r),
    ("8 (Pump out)", T8_r, P_high2, h8_r, s8_r),
]
for name, T, P, h, s in rows2:
    print(f"{'name':^14} | {T-273.15:^9.2f} | {P/1e6:^9.3f} | {h/1e3:^11.2f} | {s/1e3:^11.2f}")
print("=" * 70)

print("\n" + "=" * 48)
print("  Problem 2.1 – Performance Summary")
print("=" * 48)
print(f"  Reheater heat input      Q_reheat  = {Q_reheat:.1f} kJ/kg")
print(f"  Boiler heat input       Q_boiler   = {Q_boiler2:.1f} kJ/kg")
print(f"  Total heat input        Q_total    = {Q_total:.1f} kJ/kg")
print(f"  Total turbine work out  W_turbines = {W_turb_total:.1f} kJ/kg")
print(f"  Pump work input         W_pump     = {W_pump2_kJ:.2f} kJ/kg")
print(f"  Net work output         W_net      = {W_net2:.1f} kJ/kg")
print(f"  Thermal efficiency      η_th       = {eta_th2:.4f} ({eta_th2*100:.1f} %)")
print(f"  Circulation rate (1 MW) ṁ          = {m_dot2:.0f} kg/h")
print("=" * 48)

```

```

=====
, Problem 2.1 – State Point Property Table
=====
,
, State | T [°C] | P [MPa] | H [kJ/kg] | S [kJ/kgK]
,-----
, 3 (Turb.I in) | 500.00 | 6.000 | 3423.11 | 6.8826
, 4' (I ideal) | 281.59 | 1.400 | 2999.98 | 6.8826
, 4 (Turb.I out) | 300.62 | 1.400 | 3042.29 | 6.9576
, 5 (Reheat out) | 500.00 | 1.400 | 3474.78 | 7.6047
, 6 (Turb.II out) | 45.81 | 0.010 | 2516.75 | 7.9384
, 7 sat L | 45.81 | 0.010 | 191.81 | 0.6492
, 8 (Pump out) | 46.37 | 6.000 | 199.37 | 0.6540
,-----
,
,=====
, Problem 2.1 – Performance Summary
=====
, Reheater heat input Q_reheat = 432.5 kJ/kg
, Boiler heat input Q_boiler = 3223.7 kJ/kg
, Total heat input Q_total = 3656.2 kJ/kg
, Total turbine work out W_turbines = 1338.9 kJ/kg
, Pump work input W_pump = 7.56 kJ/kg
, Net work output W_net = 1331.3 kJ/kg
, Thermal efficiency  $\eta_{th}$  = 0.3641 (36.41%)
, Circulation rate (1 MW)  $\dot{m}$  = 2704 kg/h
,=====

```

```

In [116]: fig, ax = plt.subplots(figsize=(10, 6.5))

ax.plot(dome['s_f'], dome['T'], 'steelblue', lw=2.5)
ax.plot(dome['s_g'], dome['T'], 'steelblue', lw=2.5, label='Saturation Dome')
ax.fill_betweenx(dome['T'], dome['s_f'], dome['s_g'], alpha=0.07, color='steelblue')

segments = [
    ([s7_r, s8_r], [T7_r, T8_r], 'royalblue', 'Pump (7→8)', '--'),
    ([s8_r, s3_r], [T8_r, T3_r], 'darkorange', 'Boiler (8→3)', '-'),
    ([s3_r, s4_r], [T3_r, T4_r], 'tomato', 'Turbine I (3→4)', '-'),
    ([s4_r, s5_r], [T4_r, T5_r], 'mediumpurple', 'Reheater (4→5)', '-'),
    ([s5_r, s6_r], [T5_r, T6_r], 'crimson', 'Turbine II (5→6)', '-'),
    ([s6_r, s7_r], [T6_r, T7_r], 'teal', 'Condenser (6→7)', '-'),
]

for s_seg, T_seg, color, label, ls in segments:
    ax.plot([s/1e3 for s in s_seg], [T-273.15 for T in T_seg],
            color=color, lw=2.5, ls=ls, label=label,
            marker='o', ms=8, markerfacecolor='white', markeredgewidth=2,
            markeredgcolor=color, zorder=5)

ax.plot([s4p_r/1e3], [T4p_r-273.15], 'g^', ms=10, zorder=6, label="4' (Ideal T")
ax.plot([s3_r/1e3, s4p_r/1e3], [T3_r-273.15, T4p_r-273.15],

```

```

        'g--', lw=1.5, alpha=0.6, zorder=4)

pts = [
    (s3_r, T3_r, '3', (0.05, 12)),
    (s4p_r, T4p_r, "4'", (-0.5, 12)),
    (s4_r, T4_r, '4', (0.1, -18)),
    (s5_r, T5_r, '5', (0.05, 12)),
    (s6_r, T6_r, '6', (0.1, -18)),
    (s7_r, T7_r, '7', (-0.5, -18)),
    (s8_r, T8_r, '8', (-0.15, 12)),
]
# =====

for s_pt, T_pt, lbl, (ds, dT) in pts:
    ax.annotate(lbl, xy=(s_pt/1e3, T_pt-273.15),
                xytext=(s_pt/1e3 + ds, T_pt-273.15 + dT),
                fontsize=11, fontweight='bold', color='#333333',
                arrowprops=dict(arrowstyle='->', color='gray', lw=1))

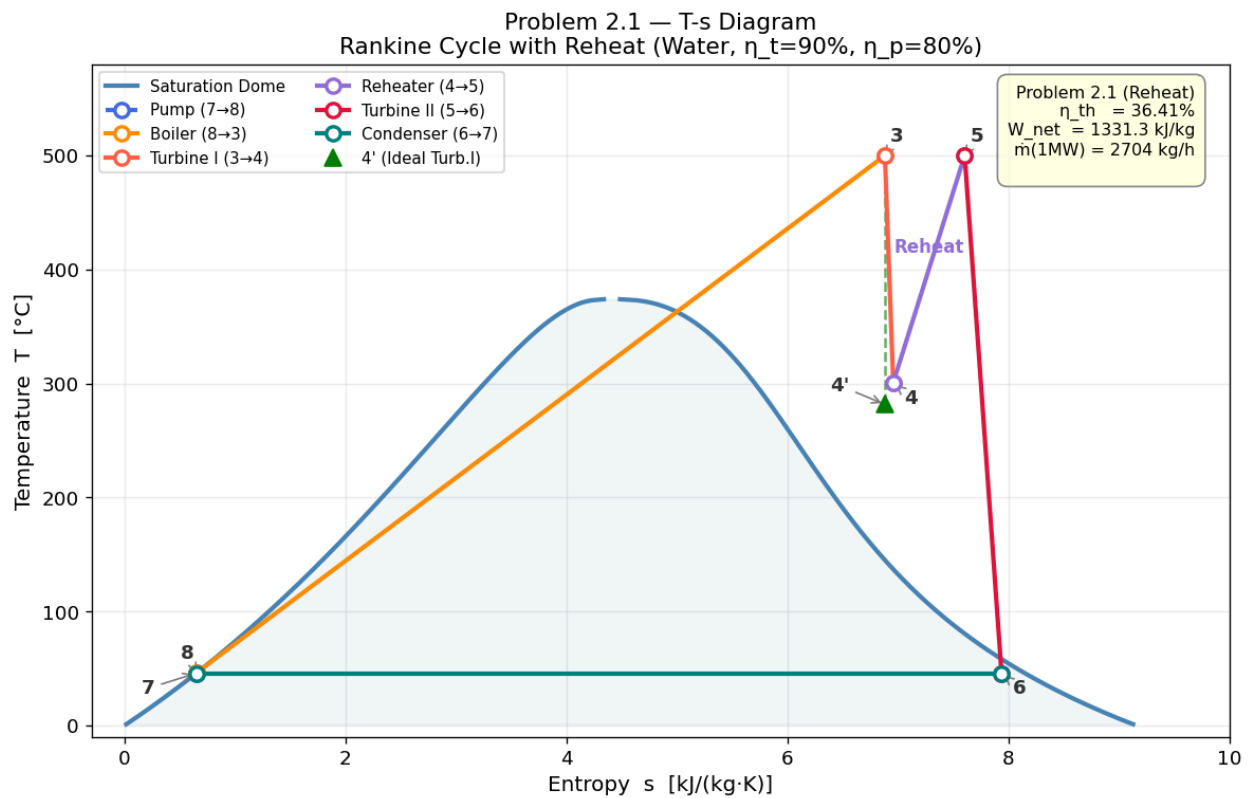
ax.annotate('Reheat', xy=((s4_r+s5_r)/2/1e3, (T4_r+T5_r)/2-273.15+15),
            fontsize=10, color='mediumpurple', fontweight='bold', ha='center')

textstr = (
    f'Problem 2.1 (Reheat)\n'
    f' $\eta_{th} = \{\eta_{th2} * 100 : .2f\} \%$ \n'
    f' $W_{net} = \{W_{net2} : .1f\}$  kJ/kg\n'
    f' $\dot{m}(1MW) = \{\dot{m}_{dot2} : .0f\}$  kg/h\n'
)
ax.text(0.97, 0.97, textstr, transform=ax.transAxes, fontsize=9.5,
        verticalalignment='top', horizontalalignment='right',
        bbox=dict(boxstyle='round,pad=0.6', facecolor='lightyellow',
                  edgecolor='gray', alpha=0.92))

ax.set_xlabel('Entropy s [kJ/(kg·K)]')
ax.set_ylabel('Temperature T [°C]')
ax.set_title('Problem 2.1 – T-s Diagram\nRankine Cycle with Reheat (Water,  $\eta_t$ ')
ax.legend(loc='upper left', fontsize=9, ncol=2)
ax.set_xlim([-0.3, 10])
ax.set_ylim([-10, 580])
ax.grid(True, alpha=0.25)

plt.tight_layout()
plt.savefig('ex52_ts_reheat.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: ex52_ts_reheat.png")

```



Saved: ex52_ts_reheat.png

Problem 3 — Basic Rankine Cycle (Three Options)

Problem Conditions

- Working fluid: **Water (Steam)**
- Turbine inlet (State 3): superheated vapor at T_3 and P_3 (varies by option)
- Condenser pressure **$P_4 = 0.01 \text{ MPa}$** for all options
- Turbine isentropic efficiency η_t and pump efficiency η_p vary by option
- Assume **no pressure change** across condenser and boiler

State Point Numbers

State 5 (sat L) → (Pump) → State 6 → (Boiler/Superheater) →
State 3
→ (Turbine) → State 4 → (Condenser) → State 5

Options (from Table 1)

Option	T3 (°C)	P3 (MPa)	P4 (MPa)	η_{turbine}	η_{pump}	Capacity
i	600	10	0.01	0.80	0.75	80 MW
ii	600	8	0.01	0.80	0.75	80 MW
iii	400	4	0.01	0.83	0.95	75 MW

Quantities to Find (per kg of steam)

- **(a)** Turbine work output — W_{turbine} [kJ/kg]
- **(b)** Pump work input — W_{pump} [kJ/kg]
- **(c)** Boiler heat input — Q_{boiler} [kJ/kg]
- **(d)** Heat-engine thermal efficiency — η_{th} [%]

```
In [120... fluid = 'Water'

options = [
    {'name': 'i', 'T3_C': 600, 'P3_MPa': 10, 'P4_MPa': 0.01,
     'eta_t': 0.80, 'eta_p': 0.75, 'cap_MW': 80},
    {'name': 'ii', 'T3_C': 600, 'P3_MPa': 8, 'P4_MPa': 0.01,
     'eta_t': 0.80, 'eta_p': 0.75, 'cap_MW': 80},
    {'name': 'iii', 'T3_C': 400, 'P3_MPa': 4, 'P4_MPa': 0.01,
     'eta_t': 0.83, 'eta_p': 0.95, 'cap_MW': 75},
]

summary = []

for opt in options:

    T3 = opt['T3_C'] + 273.15
    P3 = opt['P3_MPa'] * 1e6
    P4 = opt['P4_MPa'] * 1e6
    eta_t = opt['eta_t']
    eta_p = opt['eta_p']

    print("=" * 72)
    print(f" Option {opt['name']} - T3={opt['T3_C']}°C "
          f"P3={opt['P3_MPa']} MPa P4={opt['P4_MPa']} MPa "
          f"η_t={eta_t:.2f} η_p={eta_p:.2f} Cap={opt['cap_MW']} MW")
```

```

print("=" * 72)

h3 = PropsSI('H', 'T', T3, 'P', P3, fluid) # J/kg
s3 = PropsSI('S', 'T', T3, 'P', P3, fluid) # J/(kg·K)

s4p = s3
h4p = PropsSI('H', 'P', P4, 'S', s4p, fluid)
T4p = PropsSI('T', 'P', P4, 'S', s4p, fluid)
x4p = PropsSI('Q', 'P', P4, 'S', s4p, fluid)

h4 = h3 - eta_t * (h3 - h4p)
T4 = PropsSI('T', 'P', P4, 'H', h4, fluid)
x4 = PropsSI('Q', 'P', P4, 'H', h4, fluid)
s4 = PropsSI('S', 'P', P4, 'H', h4, fluid)

h5 = PropsSI('H', 'P', P4, 'Q', 0, fluid)
s5 = PropsSI('S', 'P', P4, 'Q', 0, fluid)
T5 = PropsSI('T', 'P', P4, 'Q', 0, fluid)
v5 = 1 / PropsSI('D', 'P', P4, 'Q', 0, fluid)

W_pump_ideal = v5 * (P3 - P4)
W_pump_actual = W_pump_ideal / eta_p

h6 = h5 + W_pump_actual
T6 = PropsSI('T', 'P', P3, 'H', h6, fluid)
s6 = PropsSI('S', 'P', P3, 'H', h6, fluid)

print(f"{'State':^13} | {'T [°C]':^9} | {'P [MPa]':^9} | "
      f"{'x':^7} | {'H [kJ/kg]':^11} | {'S [kJ/kgK]':^12}")
print("-" * 72)

rows = [
    ("3 (Turb.in)", T3, P3, 'supV', h3, s3),
    ("4 (Ideal)", T4p, P4, x4p, h4p, s4p),
    ("4 (Actual)", T4, P4, x4, h4, s4),
    ("5, sat L", T5, P4, 0.0, h5, s5),
    ("6 (Pump out)", T6, P3, 'compL', h6, s6),
]

for name, T, P, x, h, s in rows:
    x_str = f"{x:.4f}" if isinstance(x, float) else x
    print(f"{'name':^13} | {T-273.15:^9.2f} | {P/1e6:^9.3f} | "
          f"{'x_str':^7} | {h/1e3:^11.3f} | {s/1e3:^12.5f}")
print("-" * 72)
print(f" v5 = {v5*1e3:.4f} L/kg ({v5:.6f} m³/kg)")

```



```

W_turb = (h3 - h4) / 1e3
W_pump = (h6 - h5) / 1e3
Q_boil = (h3 - h6) / 1e3
Q_cond = (h4 - h5) / 1e3
W_net = W_turb - W_pump
eta_th = W_net / Q_boil
BWR = W_pump / W_turb

m_dot = opt['cap_MW'] * 1e6 / (W_net * 1e3)

print(f"\n (a) Turbine work output W_turbine = {W_turb:.3f} kJ/kg")
print(f" (b) Pump work input W_pump = {W_pump:.4f} kJ/kg")
print(f" (c) Boiler heat input Q_boiler = {Q_boil:.3f} kJ/kg")
print(f" Condenser heat reject Q_cond = {Q_cond:.3f} kJ/kg")
print(f" Net work W_net = {W_net:.3f} kJ/kg")
print(f" (d) Thermal efficiency η_th = {eta_th*100:.3f}%")
print(f" Back-work ratio BWR = {BWR*100:.3f}%")
print(f" Mass-flow ({opt['cap_MW']} MW) ṁ = {m_dot:.4f} kg/s")
print()

summary.append((opt['name'], W_turb, W_pump, Q_boil, eta_th * 100))

print("=" * 80)
print(" PROBLEM 3 – SUMMARY TABLE")
print("=" * 80)
hdr = (f"{'Option':^8} | {'a: W_turb':^16} | {'b: W_pump':^16} | "
      f"{'c: Q_boil':^16} | {'d: η_th':^12}")
print(hdr)
print(f"{'':^8} | {'[kJ/kg]':^16} | {'[kJ/kg]':^16} | "
      f"{'[kJ/kg]':^16} | {'[%]':^12}")
print("-" * 80)
for name, Wt, Wp, Qb, eta in summary:
    print(f" {name:^6} | {Wt:^16.3f} | {Wp:^16.4f} | {Qb:^16.3f} | {eta:^12.3f}")
print("=" * 80)

```

```

=====
, Option i - T3=600°C P3=10 MPa P4=0.01 MPa  $\eta_t=0.80$   $\eta_p=0.75$  Cap=80 MW
=====
, State | T [°C] | P [MPa] | x | H [kJ/kg] | S [kJ/kgK]
-----
, 3 (Turb.in) | 600.00 | 10.000 | supV | 3625.758 | 6.90447
, 4' (Ideal) | 45.81 | 0.010 | 0.8341 | 2186.967 | 6.90447
, 4 (Actual) | 45.81 | 0.010 | 0.9544 | 2474.725 | 7.80666
, 5, sat L | 45.81 | 0.010 | 0.0000 | 191.806 | 0.64920
, 6 (Pump out) | 46.95 | 10.000 | compL | 205.263 | 0.65979
-----
, v5 = 1.0103 L/kg (0.001010 m³/kg)
,
, (a) Turbine work output W_turbine = 1151.033 kJ/kg
, (b) Pump work input W_pump = 13.4568 kJ/kg
, (c) Boiler heat input Q_boiler = 3420.495 kJ/kg
, Condenser heat reject Q_cond = 2282.919 kJ/kg
, Net work W_net = 1137.576 kJ/kg
, (d) Thermal efficiency  $\eta_{th}$  = 33.258%
, Back-work ratio BWR = 1.169%
, Mass-flow (80 MW)  $\dot{m}$  = 70.3250 kg/s
,
=====
, Option ii - T3=600°C P3=8 MPa P4=0.01 MPa  $\eta_t=0.80$   $\eta_p=0.75$  Cap=80 MW
=====
, State | T [°C] | P [MPa] | x | H [kJ/kg] | S [kJ/kgK]
-----
, 3 (Turb.in) | 600.00 | 8.000 | supV | 3642.378 | 7.02211
, 4' (Ideal) | 45.81 | 0.010 | 0.8498 | 2224.487 | 7.02211
, 4 (Actual) | 45.81 | 0.010 | 0.9683 | 2508.065 | 7.91119
, 5, sat L | 45.81 | 0.010 | 0.0000 | 191.806 | 0.64920
, 6 (Pump out) | 46.72 | 8.000 | compL | 202.569 | 0.65766
-----
, v5 = 1.0103 L/kg (0.001010 m³/kg)
,
, (a) Turbine work output W_turbine = 1134.313 kJ/kg
, (b) Pump work input W_pump = 10.7628 kJ/kg
, (c) Boiler heat input Q_boiler = 3439.809 kJ/kg
, Condenser heat reject Q_cond = 2316.259 kJ/kg
, Net work W_net = 1123.550 kJ/kg
, (d) Thermal efficiency  $\eta_{th}$  = 32.663%
, Back-work ratio BWR = 0.949%
, Mass-flow (80 MW)  $\dot{m}$  = 71.2029 kg/s
,
=====
, Option iii - T3=400°C P3=4 MPa P4=0.01 MPa  $\eta_t=0.83$   $\eta_p=0.95$  Cap=75 MW
=====
, State | T [°C] | P [MPa] | x | H [kJ/kg] | S [kJ/kgK]
-----
, 3 (Turb.in) | 400.00 | 4.000 | supV | 3214.473 | 6.77136
, 4' (Ideal) | 45.81 | 0.010 | 0.8163 | 2144.508 | 6.77136
, 4 (Actual) | 45.81 | 0.010 | 0.8924 | 2326.402 | 7.34163
, 5, sat L | 45.81 | 0.010 | 0.0000 | 191.806 | 0.64920

```

,6 (Pump out) | 45.99 | 4.000 | compL | 196.049 | 0.64987

, v5 = 1.0103 L/kg (0.001010 m³/kg)

, (a) Turbine work output W_turbine = 888.071 kJ/kg
, (b) Pump work input W_pump = 4.2431 kJ/kg
, (c) Boiler heat input Q_boiler = 3018.424 kJ/kg
, Condenser heat reject Q_cond = 2134.596 kJ/kg
, Net work W_net = 883.828 kJ/kg
, (d) Thermal efficiency η_{th} = 29.281%
, Back-work ratio BWR = 0.478%
, Mass-flow (75 MW) $\dot{m}$ = 84.8582 kg/s

==
, PROBLEM 3 – SUMMARY TABLE

==
, Option | a: W_turb | b: W_pump | c: Q_boil | d: η_{th}
, | [kJ/kg] | [kJ/kg] | [kJ/kg] | [%]
, -----
, --
, i | 1151.033 | 13.4568 | 3420.495 | 33.258
, ii | 1134.313 | 10.7628 | 3439.809 | 32.663
, iii | 888.071 | 4.2431 | 3018.424 | 29.281

Problem 1-2.

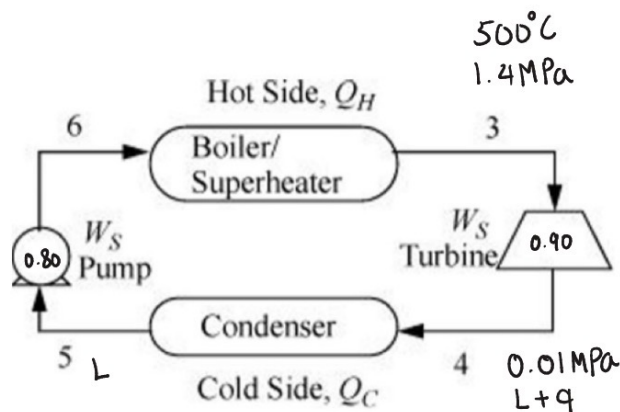


Figure 1. Rankine cycle

$$a) \frac{\dot{W}_{S, \text{turb, rev}}}{\dot{m}} = \hat{H}_{\text{out}} - \hat{H}_{\text{in}}$$

$$x = \frac{\hat{S} - \hat{S}^L}{\hat{S}^V - \hat{S}^L} = \frac{7.6047 - 0.6492}{8.1488 - 0.6492} = 0.927$$

$$\hat{H}_{\text{out}} - \hat{H}_{\text{in}} = [191.8 + (0.927)(2583.9 - 191.8)] - 3474.8 = -1065.52 \frac{\text{kJ}}{\text{kg}}$$

$$0.90 = \frac{\dot{W}_{\text{act, t}}}{-1065.52} \Rightarrow \frac{\dot{W}_{\text{act, t}}}{\dot{m}} = \boxed{-958.97 \frac{\text{kJ}}{\text{kg}}}$$

$$b) \frac{\dot{W}_{\text{res, p}}}{\dot{m}} = \hat{V} (P_{\text{out}} - P_{\text{in}}) = (0.001010)(14 - 0.1)(100) = 1.4039 \frac{\text{kJ}}{\text{kg}}$$

$$\frac{\dot{W}_{\text{act, p}}}{\dot{m}} = \frac{1.4039}{0.80} = \boxed{1.75 \frac{\text{kJ}}{\text{kg}}}$$

$$\frac{\dot{Q}_C}{\dot{m}} = \hat{H}_{\text{out}} - \hat{H}_{\text{in}}$$

$$\frac{\dot{W}_{\text{act, t}}}{\dot{m}} = \hat{H}_{\text{out, act}} - \hat{H}_{\text{in}} \Rightarrow \hat{H}_{\text{out, act}} = 2515.83 \frac{\text{kJ}}{\text{kg}}$$

$$\Rightarrow \frac{\dot{Q}_C}{\dot{m}} = 191.8 - 2515.83 = -2324.03 \frac{\text{kJ}}{\text{kg}}$$

$$\Rightarrow \frac{\dot{Q}_H}{\dot{m}} = 2324.03 - 1.75 + 958.97 = \boxed{3281.25 \frac{\text{kJ}}{\text{kg}}}$$

$$\eta_{\text{HE}} = \frac{\dot{W}_{\text{net}}}{\dot{Q}_H} = \frac{957.22}{3281.25} = \boxed{0.292 \text{ or } 29.2\%}$$

c).

Stream #	T (°C)	P (MPa)	q	H (kJ/kg)	S (kJ/kg·K)
3	500	1.4	Sup V	3474.8	7.6047
4'	45.81	0.01	0.927	2409.3	
4	45.81	0.01	0.972	2515.8	
5, sat L	45.81	0.01	0	191.8	
6		1.4	Comp L	193.6	

Note: State 4' represents the reversible process in the turbine.

$$d.) \dot{m} = \frac{1000 \text{ kW}}{957.22 \frac{\text{kJ}}{\text{kg}}} = 1.04 \frac{\text{kg}}{\text{s}}$$

Problem 2-2.

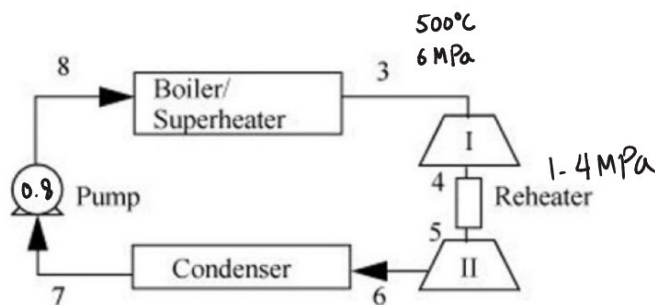


Figure 2. Rankine cycle with reheat

a).

State	T (°C)	P (MPa)	H (kJ/kg)	S (kJ/kg·K)
3	500	6	3423.1	6.8826
4'		1.4	3001.2	6.8826
4		1.4	3043.4	
7, sat L	45.81	0.01	191.8	

Note: State 4' represents the reversible process in the turbine.

$$b.) \frac{\dot{W}_{s,act,t1}}{\dot{m}} = \hat{H}_{out} - \hat{H}_{in} = 3043.3 - 3423.1 = -379.8 \frac{\text{kJ}}{\text{kg}}$$

$$\frac{\dot{W}_{s,act,t2}}{\dot{m}} = -958.97 \frac{\text{kJ}}{\text{kg}}$$

$$\dot{W}_{s,act,t} = -379.8 - 958.97 = -1338.77 \frac{\text{kJ}}{\text{kg}}$$

$$\dot{W}_{p,rev} = \hat{V}(P_{out} - P_{in}) = 0.00101(60 - 0.1)(100) = 6.05 \frac{\text{kJ}}{\text{kg}}$$

$$\dot{W}_{p,act} = \frac{6.05}{0.8} = 7.56 \frac{\text{kJ}}{\text{kg}}$$

$$\frac{\dot{Q}_c}{\dot{m}} = 191.8 - 2515.83 = -2324.03 \frac{\text{kJ}}{\text{kg}}$$

$$\frac{\dot{Q}_H}{\dot{m}} = 2324.03 - 7.56 + 1338.77 = 3655.24 \frac{\text{kJ}}{\text{kg}}$$

$$\eta_{HE} = \frac{1331.21}{3655.24} = 0.364 \text{ or } 36.4\%$$

$$\dot{m} = \frac{1000 \text{ kW}}{1331.21 \frac{\text{kJ}}{\text{kg}}} = 0.751 \frac{\text{kg}}{\text{s}}$$

d). Reheat improves thermal efficiency by increasing the average temperature of heat addition, which aligns with the Second Law of Thermodynamics to maximize the temperature gradient between the source and sink